General Segmentation Pipeline Development for Microscopy Images with Complex Cell Morphologies

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ABSTRACT

Existing tools for automated segmentation of microscopy images often require programming expertise, are hardware-restricted, or fail to generalise across diverse cell types and imaging conditions. This work makes two contributions: a quantitative comparison of five segmentation algorithms spanning classical image processing, deep learning, and the state-of-the-art Cellpose-SAM hybrid architecture as well as the development of a novel training-free pipeline combining Sobel edge detection with a marker-controlled Watershed transform. All algorithms were evaluated on an out-of-distribution dataset of densely packed mouse thymus cells using four complementary metrics assessing pixel-level overlap and boundary localisation, followed by qualitative evaluation on six fibroblast and two microglial image sets spanning varied contrast, sharpness, and cell density. Cellpose-SAM achieved the strongest quantitative performance, confirming training dataset diversity as the primary driver of deep learning generalisation. The proposed Sobel-Watershed pipeline achieved second-best overall performance, completed segmentation in approximately one second on a standard laptop, and outperformed Cellpose-SAM on microglial cells, demonstrating that deep learning models can fail on morphologies absent from training data, and that well-designed classical approaches can surpass them in such regimes. A sustainability analysis estimated the carbon cost of training Cellpose-SAM at approximately $50\,\mathrm{kg\,CO_2}$, versus $2.6\,\mathrm{kg}$ for the full classical pipeline development, underscoring the environmental case for classical methods. All code is publicly available and integrated into an existing graphical user interface, enabling researchers without programming expertise to apply and compare segmentation methods directly.

Keywords: microscopy image segmentation, Sobel-Watershed transform, deep learning, Cellpose-SAM, generalisation

NOMENCLATURE

CNN Convolutional Neural Network
DWT Discrete Wavelet Transform
Mask R-CNN Mask Region-based CNN architecture used for segmentation in the open-source MATLAB library
S+W Sobel-Watershed segmentation algorithm developed in this work

1 INTRODUCTION

Modern microscopes can capture more than 100,000 images per day [1]. High-throughput imaging is now routine in the study of cellular responses to environment and mechanical stimuli, as demonstrated by the ongoing work of the UCL Ultrasonics Group investigating microglial morphological responses to ultrasound. Extracting quantitative insight from these datasets requires accurate delineation of individual cells, a process known as segmentation, which must be automated to be tractable at scale.

Microscopy images vary substantially in cell density, size, contrast, sharpness, and imaging modality, with bright-field, fluorescence, and phase-contrast microscopy each producing images of fundamentally different character. Tools such as ImageJ require significant parameter refinement and domain expertise, and their semi-automatic workflows are impractical for large datasets [2]. Proprietary solutions such as Sartorius IncuCyte automate the process but are restricted to compatible hardware [3]. The prevailing research practice of developing bespoke pipelines for each dataset demands software engineering expertise that presents a considerable barrier in a typical biological laboratory [1].

Recent advances in machine learning, and particularly Convolutional Neural Networks (CNNs), have enabled algorithms that infer segmentation rules directly from data rather than relying on manually defined processing steps [4]. The latest generation of models integrates CNN architectures with transformer-based encoders, substantially improving generalisation across imaging conditions [5]. Nevertheless, even state-of-the-art models struggle when applied to cell morphologies absent from their training data [6], hence perfect segmentation of densely packed or overlapping cells remains an open challenge [7]. Critically, no prior work has provided a rigorous quantitative comparison between classical and machine learning approaches, leaving the assumption that machine learning invariably outperforms classical methods unsubstantiated.

This work addresses these gaps with three objectives: to conduct a rigorous comparative study of segmentation algorithms spanning classical, machine learning, and deep learning approaches, serving as a quantitative guide for future methodological decisions; to develop an accessible, training-free pipeline based on Sobel edge detection and the Watershed transform; and to make all code publicly available via GitHub, enabling researchers without programming expertise to apply the tools in alignment with the FAIR Data Principles [8].

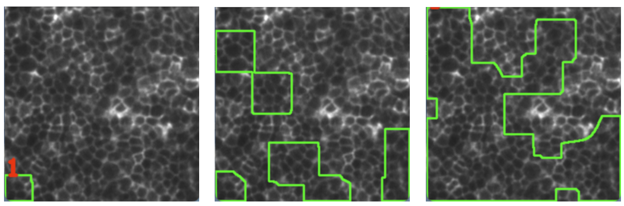
2 LITERATURE REVIEW

An extensive literature review identified a considerable number of segmentation algorithms, where the majority were developed and validated in isolation for specific datasets or cell types. Only one study directly compared multiple algorithms [6], and even this was limited to deep learning approaches. The present review evaluates four established approaches alongside the Sobel-Watershed (S+W) algorithm developed in this work. Each method was fine-tuned prior to evaluation to ensure a fair comparison at peak performance, with initial quantitative assessment conducted on a common Base image (see Section 3.1).

2.1 DWT-Based Pre-processing with Intensity Thresholding Segmentation

This pipeline was developed to automate cell image analysis for Retinal Pigment Epithelium (RPE) cells [9]. The pre-processing stage applies histogram equalisation to improve dynamic range, followed by Discrete Wavelet Transform (DWT) denoising, which decomposes the image into frequency components and suppresses high-frequency noise while preserving structural features relevant to segmentation [10]. Segmentation is then performed by binarising the denoised image using a global intensity threshold $T = \mu - \frac{1}{2}\sigma$, where μ and σ are the mean and standard deviation of pixel intensities respectively. This is followed by morphological operations to remove artefacts and consolidate boundaries, achieving 95.86% cell counting accuracy on the RPE dataset. A key limitation is that tightly packed or overlapping cells are identified as clusters rather than individual instances.

Fine-Tuning. The DWT wavelet function and decomposition level were optimised using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) [11]. Higher decomposition levels achieve greater noise suppression but halve spatial resolution at each level, creating a trade-off that must be balanced carefully. As shown in Figure 1, 3 levels produced too few detections and 5 levels identified the entire image as a single region, hence 4 levels were selected as the optimal compromise. Among the wavelet families evaluated, `bior5.5` achieved the best balance of PSNR (35.8 dB) and SSIM (0.895) and was adopted. The kernel size was confirmed at 25×25 pixels, consistent with the original configuration.



(a) 3 levels **(b)** 4 levels **(c)** 5 levels

Figure 1: Effect of decomposition level on segmentation output.

2.2 Mask R-CNN Deep Learning Segmentation with DWT Pre-processing

This pipeline is an open-source MATLAB library [3], presented at the 17th Hamlyn Symposium on Medical Robotics [12], providing end-to-end cell segmentation via DWT-based pre-processing and a Mask R-CNN architecture, with a GUI for accessibility.

Pre-processing follows a similar approach to the previous algorithm but uses soft thresholding: rather than zeroing all detail coefficients, only those below a specified percentage of the maximum coefficient magnitude are suppressed, preserving greater structural detail and avoiding artificial discontinuities [12]. Segmentation is performed using Mask R-CNN [13], a two-stage architecture that first proposes candidate regions and then produces class labels, bounding boxes, and binary segmentation masks in parallel. This enables detection of adjacent and overlapping cells, which the thresholding-based method could not achieve.

Fine-Tuning. For pre-processing, 3 decomposition levels at a 1% soft threshold yielded 180 detected cells versus 167 at default settings. The lower threshold was chosen to preserve high-frequency structural detail given the high cell confluency of the Base image. For segmentation, the mask overlap threshold was increased from 30% to 50%, yielding 5 additional detections without double-counting; the proposal overlap threshold was set to 35%, as higher values introduced double-counting and lower values caused missed detections. The EfficientNet backbone was used exclusively, as ResNet50 achieved only 2 detections and ResNet101 identified none. A noted limitation of the GUI implementation, that segmentation applies to a resized image rather than original, causing detections to extend beyond true boundaries, was subsequently addressed in Section 3.3.3.

2.3 Deep Learning + Transformer based Cellpose-SAM

Cellpose-SAM [5] is the most recent iteration of the Cellpose family of algorithms and represents the current state-of-the-art for automated cell segmentation in biomedical microscopy [14]. It integrates the pre-trained Vision Transformer (ViT-L) backbone of the Segment Anything Model (SAM) [15] — a foundation model trained on over 1 billion image annotations — with the Cellpose segmentation framework, combining the strong generalisation capacity of a large-scale transformer with a biologically motivated segmentation methodology.

Transformer-based architectures have increasingly supplanted CNNs across modern computer vision tasks [16]. Rather than applying fixed convolutional filters locally, transformers divide an image into patches and process them via a self-attention mechanism, dynamically learning relationships between distant image regions, capturing global contextual information [17], which makes them well-suited to segmenting cells of varying sizes, shapes, and textures. The known drawbacks are high computational cost and large data requirements for pretraining [17]. In hybrid architectures such as Cellpose-SAM, the transformer encoder handles global feature extraction and coarse segmentation, whilst deep learning convolutional layers refine predictions locally to capture fine-grained structural details such as cell boundaries.

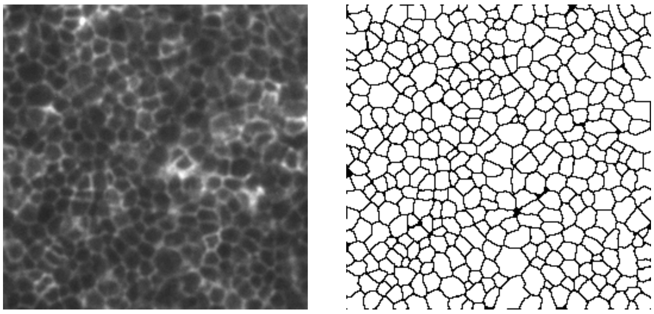
The underlying Cellpose segmentation approach [18] reformulates cell detection as a gradient flow problem: a neural network predicts, for each pixel, the probability of belonging to a cell interior and a vector pointing toward the cell centre, with pixels grouped by following these vectors to a common centre (gradient tracking). This avoids the monotonic intensity gradient assumption of watershed methods, which frequently fails in practice and causes over-segmentation [18]. Cellpose-SAM fine-tunes the SAM encoder on 22,826 microscopy images comprising over 3.3 million annotated regions [5], building on image restoration network Cellpose3 [19], which established that training data diversity is the primary determinant of generalisation. Benchmarking against alternative algorithms showed that on a LIVE-Cell dataset [20], Cellpose-SAM achieved a minimum Average Precision of 0.86 at IoU = 0.5, compared to 0.48 for the Single-Cell Transformer Segmenter (another hybrid architecture) [21] — an improvement of approximately 79% [5]. A comparative study identified Cellpose3 — the direct predecessor to Cellpose-SAM — as the best-performing method for densely packed cells, outperforming SAM and MEDIAR (another transformer based architecture) [22] on unseen data [6], with Cellpose-SAM providing a further performance boost.

No fine-tuning was required; the algorithm was run using the pre-trained `cyto3` model with default settings, consistent with the conservative approach recommended by the authors [19]. The framework also supports human-in-the-loop retraining [23], enabling periodic updates on increasingly diverse cell types, and provides a GUI for one-click segmentation.

3 METHODOLOGY

3.1 Base Image

All algorithms were evaluated quantitatively against a common reference: the Base Image (Figure 2), sourced from the CellBinDB dataset [6]. It is a large-scale benchmark of over 1,000 annotated microscopy images spanning 30+ tissue types and four staining modalities, with ground truth produced through semi-automatic and manual annotation verified by expert annotators. All experimental protocols adhered to ethical regulations regarding animal research. The selected image depicts mouse thymus tissue stained via multiplex immunofluorescence (mIF), presenting a challenging segmentation scenario of densely packed cells of varying sizes — conditions known to significantly reduce automated



(a) Base Image **(b)** Ground truth

Figure 2: Chosen samples from the CellBinDB dataset [6].

segmentation accuracy [6]. Critically, no algorithm assessed was trained on CellBinDB data, meaning all results reflect true out-of-distribution generalisation performance, directly addressing the gap identified in the Literature Review.

3.2 Segmentation Evaluation Metrics

Overlap-based metrics alone (such as F1 score) are insufficient for segmentation evaluation, as they measure pixel-level agreement without accounting for the spatial position of errors [24]. This is particularly important where accurate boundary delineation is essential for morphological analysis. Four complementary metrics were therefore employed, combining overlap and distance-based approaches.

F1 Score. Precision and recall quantify the proportion of correctly identified foreground pixels (quality) and the proportion of true foreground pixels detected (quantity), respectively:

$$\text{Precision} = \frac{TP}{TP+FP}, \ \text{Recall} = \frac{TP}{TP+FN} \tag{1}$$

where TP are true positives (correctly predicted foreground pixels), FP false positives (background pixels incorrectly predicted as foreground), and FN false negatives (foreground pixels missed by the algorithm). The F1 score is their harmonic mean [6], ranging from 0 to 1 (perfect overlap), with $\geq$0.85 generally considered strong performance in biological segmentation:

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Boundary F1 (BF) Score. The BF score applies the same precision–recall framework but only to boundary pixels, counting a predicted boundary pixel as a true positive only if a ground truth boundary pixel exists within a spatial tolerance τ [25]. A tolerance is necessary as annotators introduce inherent uncertainty in boundary placement [26]. MATLAB default of 0.75% of the image diagonal was adopted, giving $\tau \approx 3$ pixels for the 256×256 Base Image. Computed using the MATLAB `bfscore` function [27].

95th Percentile Hausdorff Distance (HD95). HD95 quantifies the magnitude of boundary deviation, computing the 95th percentile of the maximum of all minimum distances from each point on one boundary to the nearest point on the other, avoiding sensitivity to outliers [24]:

$$HD_{95}(A,B) = x_{0.95}\left\{\sup_{a\in A} d(a,B), \sup_{b\in B} d(b,A)\right\}$$

where A and B are the sets of boundary pixels of the predicted and ground truth masks respectively, $d(\cdot,\cdot)$ denotes the minimum Euclidean distance from a point to a set, sup the supremum, and $x_{0.95}$ the 95th percentile of the combined directed distances. Reported in pixels.

Average Symmetric Surface Distance (ASSD). Where HD95 captures the worst-case boundary deviation (excluding outliers), ASSD quantifies the average deviation across all boundary points in both directions [24]:

$$ASSD(A,B) = \frac{1}{|A|+|B|}\left(\sum_{a\in A} d(a,B) + \sum_{b\in B} d(b,A)\right)$$

ASSD provides a more holistic measure of boundary quality than HD95, as it is not dominated by any single region of mismatch. A value of 0 indicates perfect correspondence. ASSD is also reported in pixels.

A MATLAB code was developed to calculate these metrics for each segmentation algorithm. A metrics summary is given in Table 1.

Table 1: Segmentation metrics (higher F1/BF and lower HD95/ASSD mean better segmentation).

Metric	Description	Range	Optimal
F1	Pixel overlap (precision–recall balance)	$[0,1]$	↑
BF	Boundary overlap within tolerance τ	$[0,1]$	↑
HD95	95th percentile boundary error (px)	$[0,\infty)$	↓
ASSD	Mean symmetric boundary error (px)	$[0,\infty)$	↓

3.3 Sobel–Watershed Segmentation Algorithm Development (S+W)

The S+W algorithm is a classical image processing pipeline designed to be broadly applicable across microscopy image types. The conceptual basis for combining Sobel edge detection with the watershed transform was drawn from [28], which proposed iterative thresholding of Sobel gradient magnitudes to guide watershed segmentation. No code was provided in that work, and the present implementation constitutes an independent realisation and substantial extension of the idea in MATLAB. The complete pipeline is illustrated in Figure 3.

Step 1: Greyscale Conversion. All input images are converted to 8-bit greyscale, mapping pixel intensities to $[0, 255]$.

Step 2: Gaussian Pre-smoothing. A Gaussian blur is applied prior to edge detection. The Sobel operator amplifies high-frequency content indiscriminately, including noise alongside genuine cell boundaries [29]; pre-smoothing suppresses this noise, reducing spurious edges that would otherwise cause over-segmentation. The three-sigma rule [30] gives a kernel size of $2\lceil 3\sigma \rceil + 1$ pixels. The standard deviation σ is the single most influential parameter in the pipeline and is discussed further in Section 3.3.1.

Step 3: Sobel Edge Detection in the Frequency Domain. Edge detection is performed using the Sobel operator in the frequency domain, where we got to via Fast Fourier Transform (FFT). The Sobel operator acts as a directional high-pass filter with orthogonal low-pass smoothing, computing image gradients using the following kernels [31]:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \; G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \tag{3}$$

with gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$, where G_x and G_y are the horizontal and vertical gradient components respectively. Frequency-domain application produces a cleaner edge response. Two implementation details are critical: the 3×3 kernels must be zero-padded to the full image size $M \times N$ to prevent circular convolution artefacts, and the padded kernel must be shifted to align its centre with the FFT origin at $(1, 1)$ for correct spatial correspondence.

Step 4: Iterative Threshold Estimation. A data-driven threshold T is estimated iteratively to separate edge pixels from non-edge pixels [28], using the following procedure:

1. Initialise T^0 as the mean intensity of G.
2. Partition into two classes: C_L (pixels with $G < T^k$) and C_H (pixels with $G \geq T^k$); compute means μ_L, μ_H. k is the iteration index.
3. Update: $T^{k+1} = \dfrac{\mu_L + \mu_H}{2}$.
4. Repeat until $|T^{k+1} - T^k| < \varepsilon$, where ε is a small tolerance approaching 0.

Pixels with gradient magnitude exceeding T are classified as reliable cell boundaries.

Step 5: Foreground Mask Generation. A foreground mask restricts watershed segmentation to regions plausibly containing cells, preventing the distance transform from generating spurious background markers in sparse images. The mask is computed using MATLAB's `imbinarize` with adaptive Otsu thresholding [32]. The `ForegroundPolarity` parameter specifies whether cells are brighter or darker than background, and `Sensitivity` (fixed at 0.3) controls threshold stringency. The mask is then refined using morphological opening and closing by reconstruction with a disk-shaped structuring element of radius 1 pixel. Reconstruction rather than standard morphological operations substantially improves foreground delineation across diverse image types (Section 3.5), particularly enabling reliable detection in low-contrast cases. Watershed basins with more than 89% of pixels outside the foreground mask are classified as background.

Step 6: Marker Generation by Distance Transform.

Without controlled initialisation, the watershed transform produces severe over-segmentation from minor gradient variations [33]. Markers are generated from the Euclidean distance transform D of the foreground mask, where each pixel value equals its distance to the nearest background pixel, placing maxima near cell centres. A morphological reconstruction approach is used [34]: D is eroded with a disk of radius `DiskSize`, shrinking all local maxima; reconstruction then restores broad genuine maxima corresponding to cell centres whilst permanently eliminating narrow noise spikes. Local maxima are extracted, minima imposed at these locations in G, and the watershed transform applied. Connected components below the 5th percentile of component areas are removed, and basins exceeding ten times the median basin area are classified as background.

A fixed percentile threshold on D was evaluated and discarded as a marker generation strategy: as shown

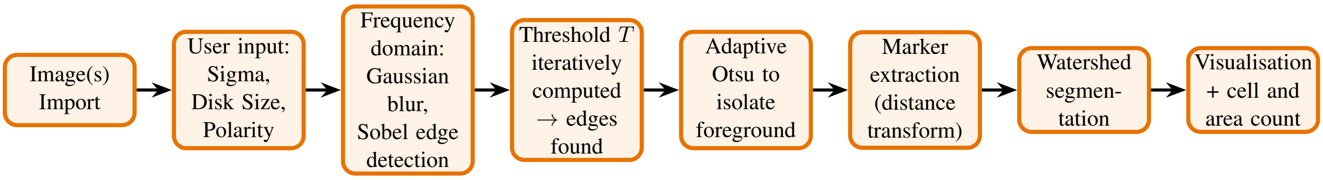


Figure 3: Flowchart of the S+W segmentation pipeline.

in Table 2, reducing the threshold from 90% to 30% found more cells and improved BF by 60.1% and halved ASSD, but at 22% produced 682 cells through severe over-segmentation. Crucially, boundary metrics continued to improve even at this threshold as a consequence of dense boundary networks reducing nearest-boundary distances. This demonstrates that boundary metrics alone cannot reliably detect over-segmentation. Such oversegmentation motivated the reconstruction approach.

Table 2: Effect of distance transform percentile threshold on segmentation metrics (Base Image). HD95 and ASSD given in pixels.

Thr.	Cells	F1	BF	HD95	ASSD
90%	17	0.8767	0.5424	20	4
70%	44	0.8643	0.6815	12	3
50%	103	0.8492	0.7897	8	2
30%	188	0.8335	0.8682	6	2
22%	682	0.7869	0.9341	3	1

Step 7: Watershed Segmentation and Output. The marker-controlled watershed transform is applied to G, flooding basins from imposed marker locations. Detected boundary pixels are rendered red in the visualisation. Cell count equals the number of non-background basins; cell area is taken as basin area in pixels.

3.3.1 Parameter Selection and Attempted Automation

The pipeline contains three user-defined parameters: Gaussian standard deviation σ, `DiskSize`, and `ForegroundPolarity`. Considerable effort was directed towards automating the calculation of these and a few more parameters from image content alone.

Gaussian Sigma. Four automation methods were investigated: linear regression on image properties (resolution, contrast, noise level estimated via the Immerkaer Laplacian method [35]); two scale-space approaches following Elder and Zucker [36] involved selecting σ at which the Sobel gradient response exceeded a noise-based significance threshold and separately finding σ at which further smoothing produced diminishing returns; and second-derivative elbow detection on the gradient response curve. All failed for the same underlying reason: σ in this pipeline compensates not only for sensor noise but also for image contrast. Low-contrast images require heavier smoothing because cell edges are weak relative to background variation, which is a fundamentally different problem from classical noise-driven sigma selection. σ therefore remains user-defined.

DiskSize. Two strategies were investigated: percentile filtering of distance transform peaks (failed because the pre-segmentation foreground mask contains merged cell regions), and a circularity-weighted approach ($4\pi \cdot \text{Area}/\text{Perimeter}^2$). A fundamental obstacle is that `DiskSize` depends simultaneously on cell size and cell separation, which vary independently and cannot both be estimated reliably from the pre-segmentation mask. This is consistent with published practice, where structuring element size is typically derived from prior knowledge of cell type and imaging conditions [37, 38]. `DiskSize` is therefore user-defined, with empirical guidance depicted in Figure 4.

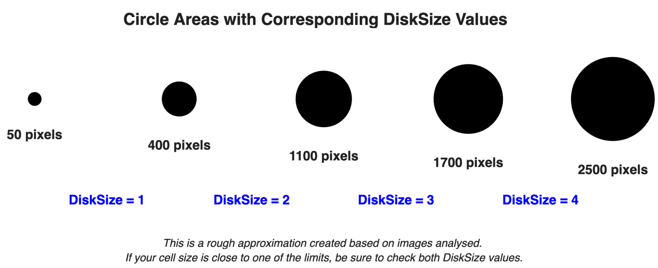


Figure 4: Guidance on DiskSize selection.

Foreground Sensitivity. An adaptive scheme was investigated in which `Sensitivity` was normalised from the local contrast of the image, measured as the mean standard deviation of pixel intensities within a sliding window of estimated cell diameter. Despite systematic evaluation across the image dataset, the computed sensitivity differed only marginally from the fixed default of 0.3 for most images, and in several high-contrast cases caused severe over-segmentation by including boundary halos in the foreground. The fixed value of 0.3 was retained. Multilevel Otsu thresholding

[32] was also investigated as an alternative foreground segmentation strategy but produced inconsistent results because it operates globally on the image histogram, making it sensitive to illumination inhomogeneities that the adaptive approach handles locally.

ForegroundPolarity. Seven automation methods were investigated, including edge-versus-non-edge intensity comparison, boundary alignment scoring, histogram skewness analysis, gradient direction sampling, and a two-stage halo detector. None generalised correctly across diverse image types due to three fundamental obstacles: structural ambiguity between cell membranes and phase contrast halos; undefined intensity comparisons in high-confluency images lacking clear background; and insufficient contrast in very low-contrast images. A robust solution would likely require a trained classifier or explicit acquisition metadata. `ForegroundPolarity` therefore remains user-defined.

Contrast Enhancement: LUTs and CLAHE. Three Look-Up Table (LUT) pre-processing strategies were evaluated (histogram matching, linear intensity scaling, and flat-field subtraction), all found unsuitable as they require calibration against a reference image, which is impractical given the diversity of the image dataset. CLAHE [39] was also evaluated, but consistently caused severe over-segmentation: local histogram equalisation amplified intra-cell texture to a degree the iterative threshold could not suppress. Both approaches were excluded from the final pipeline.

3.3.2 Pipeline Improvement

Quantitative results before and after pipeline refinement are summarised in Table 3. Pipeline refinement produced substantial improvements in all boundary-sensitive metrics: BF increased by 71.6%; HD95 decreased by 71.4%, and ASSD by 60.0%. The modest F1 decrease (5.2%) is attributable to the increase in detected cell count, as more, smaller basins reduce pixel-level overlap whilst improving boundary localisation.

3.3.3 Integration of S+W into the MATLAB GUI Application

The S+W algorithm was integrated into the existing MATLAB GUI described in Section 2.2, enabling direct comparison of the classical and deep learning pipelines within a unified interface. A dedicated S+W input

Table 3: S+W segmentation performance on the Base Image before and after pipeline refinement.

Metric	Initial	Final
Cells	13	192
F1	0.8789	0.8328
BF	0.5094	0.8740
HD95 (px.)	21	6
ASSD (px.)	5	2

panel was added to the Settings section (Figure 5), through which the user specifies σ, `DiskSize`, and `ForegroundPolarity` — the only three parameters required, in contrast to the considerably larger parameter set of the deep learning networks. The watershed output was converted into the format expected by downstream modules, ensuring all visualisation, export, and feature extraction functionality operates identically regardless of algorithm selected. A further modifica-

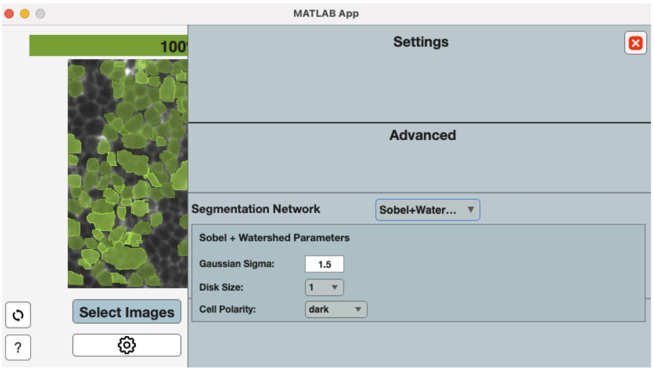


Figure 5: Snapshot of the S+W panel from the MATLAB app

tion addressed a limitation of the original application in which all algorithms operated on a resized image. Segmentation is now performed at native resolution, with resizing applied exclusively to the displayed result, so exported masks correspond to the original image resolution as well. The updated application has been published on the Ultrasonics Group GitHub repository, making the S+W algorithm accessible to researchers through a GUI without requiring any programming expertise. Images of the updated GUI can be found in Appendix 10.

3.4 Comparison Against ImageJ

ImageJ is an open-source image analysis platform that has become the de facto standard for scientific image processing in the biological sciences [2], making it a meaningful baseline against which the S+W pipeline can be compared. Despite its broad adoption, accurate segmentation requires significant parameter refinement and expertise. Its Cell Counter plugin, which is widely regarded as the benchmark tool for semi-automated segmentation tasks [40], requires manual cell marking, making it impractical for large datasets.

The ImageJ pipeline was configured to replicate the S+W processing steps as closely as possible: 8-bit greyscale conversion; Gaussian pre-smoothing with the same σ as S+W; FFT followed by *Apply Edges* to apply the Sobel operator in the frequency domain; inverse FFT; a further Gaussian blur ($\sigma = 0.5$ px) to merge broken edges; IsoData thresholding — an iterative method conceptually equivalent to the S+W threshold estimator [41]; watershed segmentation on the Euclidean distance map; and finally *Analyse* $\rightarrow$ *Analyse Particles* to produce a binary output consistent with the ground truth format.

3.5 Qualitative Evaluation on Diverse Cell Datasets

Quantitative evaluation on the Base Image is limited to a single cell type and imaging condition. To assess broader generalisability, qualitative evaluation was conducted on two additional datasets spanning distinct cell morphologies, imaging modalities, and acquisition conditions. As no ground truth annotations are available, evaluation is visual, reflecting the realistic condition in which segmentation tools are deployed in practice.

3.5.1 Fibroblast Cells

Six microscopy images of fibroblast cells (F1–F3, F5–F6 brightfield; F4 phase contrast) were provided by the UCL Ultrasonics Group, acquired using a Leica Thunder inverted light microscope at $20\times$ magnification. The dataset spans considerable variation in contrast, sharpness, and cell density, making it well suited for assessing generalisability. Two image quality metrics were computed to characterise these differences.

Contrast was measured using RMS Contrast [42], defined as the standard deviation of the normalised pixel intensities:

$$C_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(I_i - \bar{I})^2} \tag{4}$$

where I_i is the intensity of pixel i, $\bar{I}$ is the mean intensity, and N is the total number of pixels. A higher value indicates greater separation between cell and background intensities.

Sharpness was quantified using Laplacian variance [43, 44], defined as the variance of the image's second-order spatial derivative:

$$S = \mathrm{Var}\big(\nabla^2 I\big)\,,\ \nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \tag{5}$$

where $\nabla^2 I$ is the Laplacian of the image I, applied as a discrete convolution with a 3×3 kernel. Sharp images produce large second-derivative responses and hence high variance; blurry images suppress high-frequency content, yielding low variance.

The computed values for all six images are presented in Table 4. These metrics are used for relative comparison within the dataset rather than as absolute quality thresholds, as no universal benchmark exists for either measure.

Cell density (confluency) was assessed visually, as it is not straightforwardly quantifiable without a ground truth segmentation. F3 and F4 represent the high-contrast, sharp end of the spectrum, F2 and F5 the low-contrast, blurry end, and F1 and F6 occupy an intermediate range. Combined with variation in cell size and confluency, these ensure the evaluation covers a realistic cross-section of imaging conditions.

Table 4: RMS Contrast and Laplacian Variance for fibroblast images F1–F6. Higher values indicate greater contrast and sharpness respectively.

Image	RMS Contrast	Laplacian Variance
F1	0.0928	0.007838
F2	0.0874	0.006690
F3	0.1562	0.010934
F4	0.1382	0.017659
F5	0.0831	0.008010
F6	0.1020	0.008091

3.5.2 Microglial Cells

Two confocal microscopy images of mouse microglial cells (M1–M2) were provided by the UCL Ultrasonics Group, acquired from 300 μm brain slices stained with an Iba1 antibody. Microglia are the resident immune cells of the central nervous system and their morphological analysis is relevant to neuroinflammatory conditions such as dementia [45]. They present a distinct segmentation challenge: highly irregular, branching morphologies with sparse distribution. This contrasts with the densely packed Base Image and confluent fibroblast cultures, providing a qualitative test under low-density, complex-morphology conditions.

4 RESULTS AND DISCUSSION

4.1 Base Image

Table 5: Segmentation results of Base Image

Segmentation	F1	BF	HD95	ASSD	CPU(s)	Cells	#
Thresholding	0.377	0.164	115	18	0.76	4	5
Mask R–CNN	0.668	0.677	14	3	17.1	189	3
S+W	0.833	0.874	6	2	1.15	192	2
ImageJ S+W	0.808	0.595	14	3	N/A	N/A	4
Cellpose-SAM	0.840	0.966	3	1	7.74	287	1

The quantitative results for all five algorithms on the Base Image are presented in Table 5. Images of the segmentation results can be found in the additional Appendix document in Appendix 1.

Cellpose-SAM achieved the strongest performance across all metrics, consistent with its designation as state-of-the-art for densely packed cell segmentation [5], with its sole practical disadvantage being processing time. The developed S+W algorithm achieved second-best overall performance (Figure 6), with a BF score only 10.5% lower than that of Cellpose-SAM and a near-identical F1. Boundary distance metrics were approximately double those of Cellpose-SAM, indicating that correct regions are identified but boundary localisation is less precise. It is important to note that performance can be improved for any specific image through fine-tuning of 3–4 additional parameters. The results reported here reflect a general configuration intended to perform consistently across diverse image types. The competitive performance of a classical pipeline against deep learning on out-of-distribution data is a meaningful finding, consistent with arguments that classical methods remain valuable where labelled training data is scarce [7]. Mask R-CNN and ImageJ S+W occupy a similar middle tier. Mask R-CNN produced a 13.7% higher BF but 17.4% lower F1 than ImageJ, indicating more accurate boundary placement but a higher rate of false detections. It is ranked third on balance due to superior boundary accuracy. Its underperformance relative to Cellpose-SAM directly illustrates the effect of training data diversity: Mask R-CNN was trained on approximately 3,200 images from a single dataset, versus 22,826 images across 19 datasets for Cellpose-SAM. As S+W clearly outperformed ImageJ, ImageJ is excluded from further analysis. The intensity thresholding pipeline performed poorest by a substantial margin, detecting only 4 cells, which is an anticipated outcome

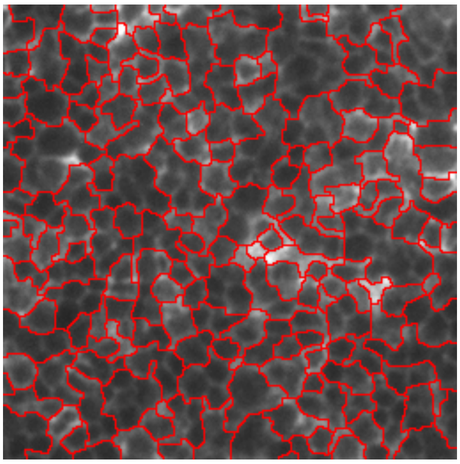


Figure 6: S+W segmentation on Base Image

given its known inability to resolve densely packed, boundary-sharing cells.

4.2 Fibroblast Cells

Visual results for F1–F6 are provided in the accompanying Appendix document (Appendix 2–7). S+W user input parameters are summarised in Table 6. The qualitative ranking from the Base Image was preserved: Cellpose-SAM outperformed all other algorithms. DWT thresholding consistently performed worst, failing to identify a single cell correctly across all six images, classifying the majority of each image as background and producing rectangular artefacts along image borders.

Table 6: S+W pipeline user input parameters used for each fibroblast and microglial image.

Image	Sigma	DiskSize	Polarity
F1	3.5	1	Darker
F2	3.0	4	Darker
F3	1.0	2	Darker
F4	3.0	1	Brighter
F5	3.1	1	Brighter
F6	2.0	2	Darker
M1	5.0	1	Brighter
M2	5.0	4	Brighter

F1 (medium contrast, blurry, high confluency): Cellpose-SAM found 849 cells, segmenting the right side well but failing on the most severely blurred left side. S+W found 726 cells with oversegmentation on the right and undersegmentation on the left, consistent with the known sensitivity of Sobel detection to blur. Mask R-CNN found 294 cells with considerable mask overlap.

F2 (blurriest and medium confluency): Cellpose-SAM found 220 cells, missing some but segmenting the majority correctly. S+W found 236 cells but misclassified background as cells due to near-identical cell and background intensities. Mask R-CNN (193 cells) performed relatively well — its best fibroblast result, likely due to this image's similarity to its LIVECell training data.

F3 (highest contrast, high confluency): S+W found 1,650 cells, comparable to Cellpose-SAM's 1,744, confirming that Sobel detection produces reliable gradients under high contrast. The main S+W artefact was occasional misidentification of multi-pixel-wide boundaries as cells — a consequence of the watershed algorithm assuming boundaries are one pixel thick. Mask R-CNN severely undersegmented (532 cells).

F4 (sharpest image, medium confluency): Cellpose-SAM (Figure 7) achieved near-perfect segmentation (1,319 cells), correctly classifying an image artefact as background. S+W (Figure 8) found 1,308 cells — remarkably close — with only minor errors. Mask R-CNN (Figure 9) found 298 cells, none correctly, apparently confused by the phase contrast halo, as was most likely trained only on images with cell interior darker than the background.

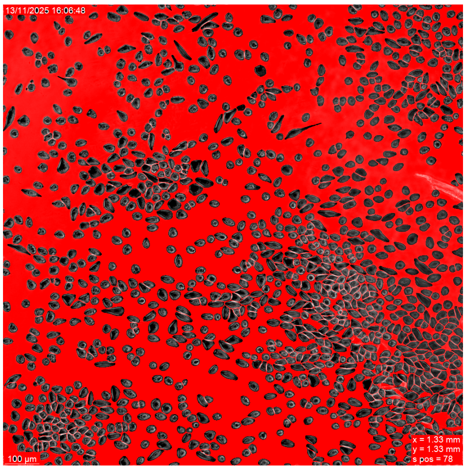


Figure 7: Cellpose-SAM segmentation of F4

F5 (lowest contrast, smallest cells): Cellpose-SAM found 184 cells with high accuracy. S+W found 82 — approximately 55% fewer — with correct boundaries on detected cells but substantial undersegmentation overall, as low contrast made foreground mask generation unreliable and the small cell size challenged the distance transform marker approach. Mask R-CNN found 38 cells, none correctly.

F6 (intermediate across all characteristics): Cellpose-SAM found 887 cells with near-perfect accuracy. S+W found 824 with minor under- and oversegmentation. Mask R-CNN found 473 cells (second-best fibroblast

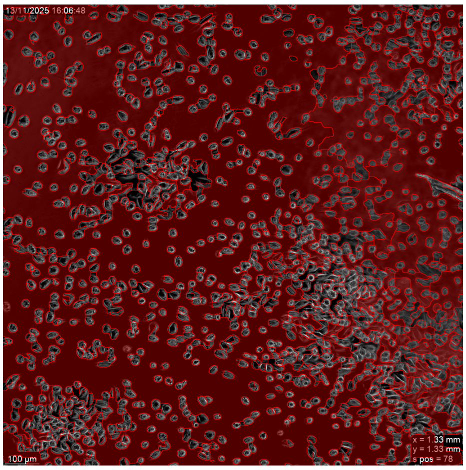


Figure 8: S+W segmentation of F4

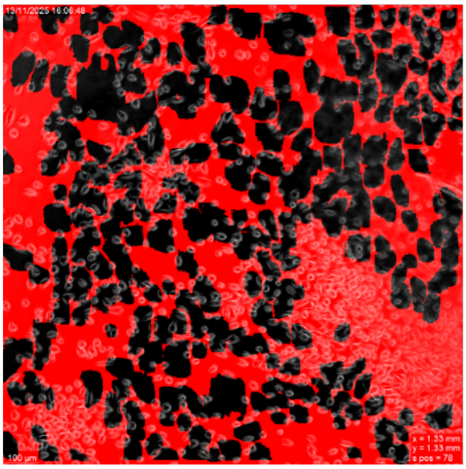


Figure 9: Mask R-CNN segmentation of F4

result) with incomplete mask coverage.

Summary. Cellpose-SAM performed consistently well, with its only notable failure on the severely blurred region of F1. S+W performed best on high-contrast images (F3, F4), produced usable results on F6, and struggled on blurry (F1, F2) and low-contrast (F5) cases. Mask R-CNN performed poorly across most images, with the inverse relationship between performance and contrast highlighting the risk of single-dataset-trained models across varied imaging conditions.

4.3 Microglial Cells

Given the poor fibroblast performance of Mask R-CNN and thresholding, only Cellpose-SAM and S+W were evaluated on the microglial images. These cells exhibit highly irregular, branching morphologies (processes) and are sparse with clear inter-cell background. They were acquired as two-dimensional cross-sections of three-dimensional structures — the greatest departure from Base Image conditions in this study.

M1 (more circular microglia, high density of small fragments): Cellpose-SAM found 163 cells, missing

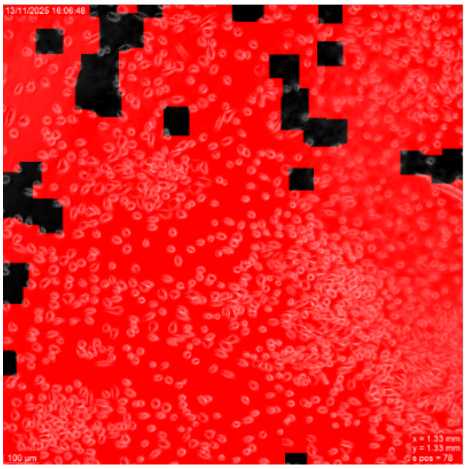
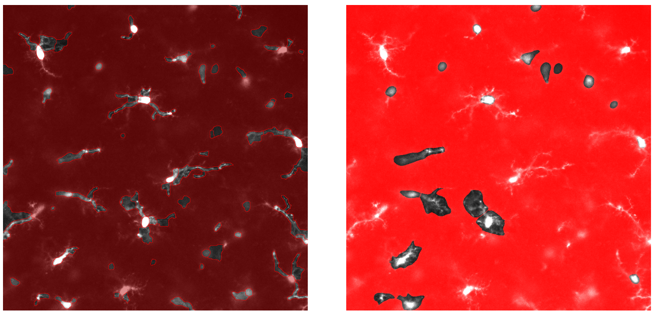


Figure 10: Thresholding segmentation of F4

approximately 60% of structures and failing on any cell with visible branching processes — suggesting circular morphology is a prerequisite for reliable detection. S+W found 792 cells, accurately segmenting the small circular fragments with minor oversegmentation of larger bodies.

M2 (pronounced branching, larger cells): As demonstrated in Figure 11, Cellpose-SAM found only 20 cells, almost exclusively identifying circular cell centres and completely failing to delineate any cellular processes. S+W found 93 cells, oversegmenting processes into multiple regions but at least detecting them — a meaningful qualitative advantage for morphological analysis.



(a) S+W result **(b)** Cellpose-SAM result

Figure 11: M2 segmentation results

Summary. The microglial dataset produced the only consistent reversal of the performance ranking: S+W outperformed Cellpose-SAM on both images. This directly illustrates the well-documented failure mode of deep learning models when applied to morphologies absent from their training data [6], and supports the argument that classical methods retain practical value precisely in such regimes [7].

4.4 Overall Discussion

Cellpose-SAM is confirmed as the strongest general-purpose algorithm evaluated, consistent with its state-of-the-art designation [14], but the microglial results are an important caveat: even the best available model fails substantially on morphologies outside its training distribution. The developed S+W algorithm occupies a practically useful position: it is competitive with Cellpose-SAM on high-contrast images, superior on microglial cells, and considerably faster across all conditions. The failure of systematic parameter automation attempts reflects genuine ambiguity in diverse microscopy image content [45, 46], and the need for user-defined parameters is not unique to classical pipelines: even state-of-the-art parameter-free models require minimum object size to be user-specified [47]. The fact that a training-free, lightweight pipeline can match the current state-of-the-art across multiple image types and surpass it on morphologically complex cells underscores its value as an accessible, computationally efficient tool for researchers lacking the data, hardware, or expertise to deploy deep learning solutions.

5 SUSTAINABILITY ANALYSIS

This section quantifies the energy and carbon cost of training deep learning segmentation models and evaluates whether the S+W algorithm represents a more sustainable alternative.

Training a state-of-the-art Cellpose-SAM [5] required eight NVIDIA H200 GPUs (Thermal Design Power 700 W each [48], combined GPU draw of 5.6 kW) for approximately 20 hours, giving a total GPU energy consumption of:

$$E_{\mathrm{GPU}} = 5.6\ \mathrm{kW} \times 20\ \mathrm{h} = 112\ \mathrm{kWh}$$

Using the global average carbon intensity of electricity generation in 2024 — 445 $\mathrm{gCO_2/kWh}$ [49], the estimated carbon footprint of a single Cellpose-SAM training run is:

$$112\ \mathrm{kWh} \times 0.445\ \mathrm{kgCO_2/kWh} \approx \mathbf{50}\ \mathrm{kgCO_2}$$

The S+W algorithm requires no training. Assuming 130 hours of active development on a MacBook Air 2023 (45 W under load):

$$E_{\mathrm{S+W}} = 0.045\ \mathrm{kW} \times 130\ \mathrm{h} = 5.85\ \mathrm{kWh} \approx \mathbf{2.6}\ \mathrm{kgCO_2}$$

The 50 $\mathrm{kgCO_2}$ figure (equivalent to driving a petrol car for 290 km [50]) represents a single training run. In practice, deep learning models generalise poorly to morphologies outside their training distribution [6], requiring retraining and fine-tuning each time a new cell type is encountered. Hence, such carbon cost is incurred repeatedly, especially in a field where research groups routinely train dedicated models for specific cell types rather than reusing existing tools [1]. S+W incurs no such cost: applied to a new cell type, it requires only minor parameter adjustment at negligible computational expense, making its carbon cost nearly 20 times lower than a single Cellpose-SAM training run. This benefit is conditional on the image type falling within the algorithm's operating range as very blurry or low-contrast images may still necessitate deep learning algorithms to ensure accurate segmentation. The results nonetheless suggest that the field would benefit from more deliberate assessment of when a training-free approach is sufficient, and from wider adoption of well-generalised community-standard models to reduce the incentive for independent retraining across overlapping use cases.

6 CONCLUSION

This work addressed two gaps in the microscopy image segmentation literature: the absence of a rigorous quantitative comparison between classical and machine learning approaches, and the lack of an accessible, training-free segmentation tool for biological laboratories. Both objectives were met, with findings that challenge the prevailing assumption that machine learning invariably outperforms classical methods.

Cellpose-SAM demonstrated the strongest overall performance, confirming that training dataset diversity is the principal driver of deep learning generalisation. Mask R-CNN, trained on a single dataset, performed substantially worse across all image types. DWT-based thresholding confirmed, also quantitatively, that intensity thresholding cannot resolve densely packed cells regardless of pre-processing quality.

The most significant finding is that the developed S+W algorithm outperformed Cellpose-SAM on microglial cells, despite requiring no training data and completing segmentation roughly 6 times quicker on a standard laptop. This exposes a structural vulnerability of deep learning segmentation that is likely to recur wherever researchers study cell types underrepresented in public training datasets. The sustainability analysis reinforces this case: S+W development produced approximately 19 times less CO_2 than a single Cellpose-SAM training run, a gap that compounds across a field where retraining for each new cell type is common practice. At the field level, the most sustainable path forward lies in wider adoption of well-generalised community-standard models alongside maintained classical baselines, reserving the energy cost of model training for cases where it is genuinely necessary.

The primary limitation of S+W is its dependence on image contrast: Sobel-based edge detection degrades on blurry and low-contrast images. Future work should investigate whether integrating DWT-based preprocessing could improve robustness on low-contrast images by suppressing noise prior to edge detection, potentially extending the algorithm's operating range without sacrificing its training-free design. Ground truth annotation of microglial images should also be prioritised to enable quantitative validation of the promising qualitative results presented here. Active learning strategies, where a model iteratively requests annotations only for the most informative samples [51], should also be considered as a direction for reducing the labelled data burden of deep learning approaches. This might offer a middle ground between the high annotation cost of fully supervised models and the parameter sensitivity of training-free pipelines.

7 CODE AVAILABILITY

The pipeline is publicly available on GitHub with documented parameter guidance: `https://github.com/rustamtoshov23/Sobel-Watershed-segmenter.git`. MATLAB GUI version can be accessed in the "Adding-S+W" branch (or main branch when added) on this GitHub page: `https://github.com/USonixGroup/Multires-ML-Microscopy.git`.

8 DECLARATIONS

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8.2 Use of AI

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